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(54) **NEURAL NETWORK CIRCUITRY HAVING FLOATING POINT FORMAT WITH ASYMMETRIC RANGE**

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(57) **ABSTRACT**

An apparatus includes circuitry for a neural network that is configured to perform forward propagation neural network operations on floating point numbers having a first n-bit floating point format. The first n-bit floating point format has a configuration consisting of a sign bit, m exponent bits and p mantissa bits where m is greater than p. The circuitry is further configured to perform backward propagation neural network operations on floating point numbers having a second n-bit floating point format that is different than the first n-bit floating point format. The second n-bit floating point format has a configuration consisting of a sign bit, q exponent bits and r mantissa bits where q is greater than m and r is less than p.

